Third, *very simple computer programs* are used. Most DSP programs are written in C, Fortran, or a similar language. However, *learning* DSP has different requirements than *using* DSP. The student needs to concentrate on the algorithms and techniques, without being distracted by the quirks of a particular language. Power and flexibility aren't important; simplicity is critical. The programs in this book are written to teach DSP in the most straightforward way, with all other factors being treated as secondary. Good programming style is disregarded if it makes the program logic more clear. For instance:

- ❏ a simplified version of BASIC is used
- ❏ line numbers are included
- ❏ the only control structure used is the FOR-NEXT loop
- ❏ there are no I/O statements

This is the simplest programming style I could find. Some may think that this book would be better if the programs had been written in C. I couldn't disagree more.

The Intended Audience

This book is primarily intended for a one year course in practical DSP, with the students being drawn from a wide variety of science and engineering fields. The suggested prerequisites are:

- ❏ A course in practical electronics: (op amps, RC circuits, etc.)
- ❏ A course in computer programming (Fortran or similar)
- ❏ One year of calculus

This book was also written with the practicing professional in mind. Many everyday DSP applications are discussed: digital filters, neural networks, data compression, audio and image processing, etc. As much as possible, these chapters stand on their own, not requiring the reader to review the entire book to solve a specific problem.

Support by Analog Devices

The Second Edition of this book includes two new chapters on *Digital Signal Processors*, microprocessors specifically designed to carry out DSP tasks. Much of the information for these chapters was generously provided by Analog Devices, Inc., a world leader in the development and manufacturing of electronic components for signal processing. ADI's encouragement and support has significantly expanded the scope of this book, showing that DSP algorithms are only useful in conjunction with the appropriate hardware.